

3 · Experiments

Datasets, baselines, metrics, and key results.

Datasets & protocol

QM9 (≤ 9 atoms) and ZINC-250k (drug-like). 500 diffusion steps, classifier-free guidance ($\lambda = 3$ on QM9, 2 on ZINC). Targeting: 100 target properties $\times$ 10 molecules, scored by MAE. Optimization (Jin et al.): corrupt for 100 steps, denoise toward the target, keep the best candidate above a Tanimoto threshold.

Baselines

FreeGress and DiGress for targeting; JT-VAE, CG-VAE, GCPN for optimization. All use a fixed-size denoiser. GrIDDD is the only one that can insert and delete atoms.

Property targeting (MAE ↓)

QM9 μ : 0.66 vs 0.74 (FreeGress). ZINC MW: 4.89 vs 8.96, roughly half the error, and without giving the model the target size in advance. Validity holds or improves in most settings.

Optimization, OOD & ablation

QED success 45.1% vs 9.4% (GCPN), about 5 $\times$. LogP improvement 2.70. Out of distribution, GrIDDD stays 35% valid at 15 atoms (trained only up to 14); DiGress drops to near zero. Switch insertion/deletion off and QED success falls 45.1 $\rightarrow$ 33.8%, so the size control is doing the work.

Table 1: Results of property targeting on QM9.				
Method	μ		HOMO	
	MAE ↓	Val. ↑	MAE ↓	Val. ↑
Unconditional	1.68 ± 0.15	91.5 %	0.95 ± 0.10	91.5 %
DiGress	0.80 ± 0.07	82.5 %	0.61 ± 0.07	91.2 %
FreeGress	0.74 ± 0.08	83.7 %	0.32 ± 0.04	90.1 %
GRIDDD	0.66 ± 0.07	79.0 %	0.37 ± 0.07	94 %

Table 1 — property targeting on QM9 (Ninniri et al., 2025).

Table 2: Results of property targeting on the ZINC-250k dataset.						
Method	LogP		QED		MW	
	MAE ↓	Val. ↑	MAE ↓	Val. ↑	MAE ↓	Val. ↑
Unconditional	1.52 ± 0.12	86.1 %	0.15 ± 0.01	86.1 %	74.16 ± 6.71	86.1 %
DiGress	0.74 ± 0.08	74.6 %	0.15 ± 0.01	85.1 %	20.92 ± 2.90	40.4 %
FreeGress	0.17 ± 0.01	84.9 %	0.04 ± 0.01	84.9 %	8.96 ± 1.93	79.7 %
GRIDDD	0.19 ± 0.02	87.9 %	0.04 ± 0.00	87.2 %	4.89 ± 0.57	84.2%

Table 2 — property targeting on ZINC-250k.

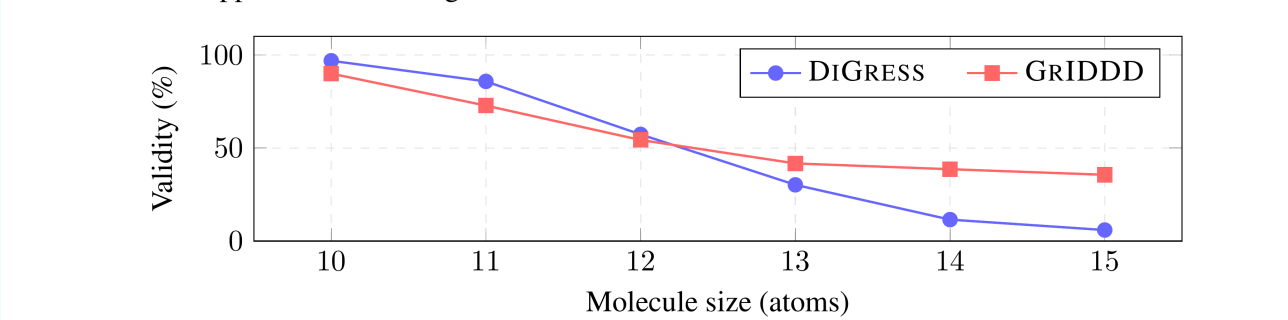


Figure 3: Validity results of out-of-distribution sampling on QM9.

Fig. 3 — out-of-distribution validity vs molecule size.

Table 3: Benchmark on graph property optimization.								
Method	LogP (sim ≥ 0.4)		LogP (sim ≥ 0.6)		QED (sim ≥ 0.4)		DRD2 (sim ≥ 0.4)	
	Improv. ↑	Div.↑	Improv. ↑	Div.↑	Succ.↑	Div.↑	Succ.↑	Div.↑
JT-VAE	1.03 ± 1.39	-	0.28 ± 0.79	-	8.8 %	-	3.4 %	-
CG-VAE	0.61 ± 1.09	-	0.25 ± 0.74	-	4.8 %	-	-	-
GCPN	2.49 ± 1.30	-	0.79 ± 0.63	-	9.4 %	0.216	4.4 %	0.152
GRIDDD	2.70 ± 0.94	0.482	1.33 ± 0.61	0.280	45.1 %	0.283	5.0 %	0.121

Table 3 — property optimization on ZINC-250k.